

Weekly Progress Report – Week 3
Intern Name: Anshika Kodukula
Organization: Trinetra Drones and Robotics Pvt.
Ltd.
Week: Week 3
Reporting Period: June 17, 2026 – June 21, 2026

Objectives

Study machine learning and deep learning models used for crop disease detection.
Analyze image classification techniques in agricultural AI systems.
Research the integration of drones, AI, and IoT in smart farming.
Develop a conceptual workflow for AI-based crop disease detection.
Compare various AI algorithms used in agriculture.
Create digital marketing content related to agricultural technology.

Activities Completed

AI Model Research

Studied Convolutional Neural Networks (CNN), ResNet, MobileNet, YOLO, Random Forest, SVM, and KNN.
Understood their applications in crop disease detection and agricultural monitoring.
Analyzed the advantages and limitations of each algorithm.

Smart Agriculture Research

Explored the integration of drones, Artificial Intelligence, and IoT technologies.
Studied precision agriculture techniques and modern farming solutions.
Reviewed real-world applications of drone-assisted crop monitoring.

Workflow Development

Designed a workflow for an AI-based Crop Disease Detection System using drone imagery.
Included image capture, preprocessing, AI analysis, disease detection, and farmer notification stages.

Comparative Analysis

Prepared a comparative analysis of AI algorithms used in agriculture.
Evaluated performance based on accuracy, speed, applications, advantages, and limitations.

Digital Marketing and Branding

Created three professional social media posts highlighting AI applications in agriculture.
Designed two technical infographics related to drone-based crop monitoring and smart farming.
Developed an educational video explaining the benefits of agricultural drones.

Skills Learned

Fundamentals of Deep Learning and Machine Learning
Image Classification Techniques
Agricultural Drone Applications
Precision Agriculture Concepts

AI-Based Crop Disease Detection
Technical Content Creation
Social Media Content Development

Deliverables Completed

AI Model Research Report
Crop Disease Detection Workflow Diagram
Comparative Analysis Report of AI Algorithms
Three Social Media Posts
Two Technical Infographics
One Educational Video

Challenges Faced

Understanding advanced deep learning architectures and their practical applications.
Comparing multiple AI algorithms based on performance metrics.
Designing technical diagrams and educational content within project timelines.

Outcomes

Developed a strong understanding of AI applications in agriculture.
Learned how drone imagery can be integrated with AI models for crop disease detection.
Improved technical research, documentation, and digital content creation skills.

Plan for Week 4

Design a conceptual AI-powered Smart Agriculture Monitoring System.
Develop a system architecture diagram integrating drones, AI, cloud platforms, and farmers.
Research future trends in agricultural drones and intelligent farming technologies.
Create promotional content and final project presentation.
Prepare the final internship report and documentation.

Conclusion

Week 3 focused on understanding AI-driven agricultural technologies, crop disease detection systems, and precision farming solutions. The activities completed during this week strengthened knowledge of machine learning, drone applications, and smart agriculture while contributing to the development of project deliverables.